Exact pairing transfer and complete cost

Frozen extension-field ECDLP targets · V2 run 2026-10-15 05:55 UTC

The public targets lie in prime-order subgroups of $E(F_{p^2})$, $E : y^2 = x^3 + 1$, with $(p, r) = (61, 19)$ and $(271, 61)$. The native solver constructed and certified $F_{p^2}\frac{[z]}{z^9-2}$ or $F_{p^2}\frac{[z]}{z^5-2}$, selected a nondegenerate r-torsion pairing input, computed the reduced Tate values, and solved the multiplicative DLP without access to the known scalar. It returned 12 and 10. Six separate source-arithmetic processes verified $[k]G = Q$; all 18 public-multiple pairing checks and 12 rho controls also passed. The answer-free input, selected H, source and binary hashes, commands, raw records and SHA-256 manifest are committed beside this report.

The one-target cold reference is same-target negation rho. The table uses $S_{\mathrm{CPU/GAE}}$ = current-thread CPU nanoseconds divided by the paired rho source-add calibration and $\sqrt{r}$. The formal two-class counted floor $\sqrt{\frac{\pi}{4}} = 0.886226925$ is a marker in this local CPU-equivalent unit, not a hardware-time bound. Each value is a three-round median; brackets show the full three-round range. Ratios to rho are medians of same-round full CPU ratios. Every answer in the table was verified.

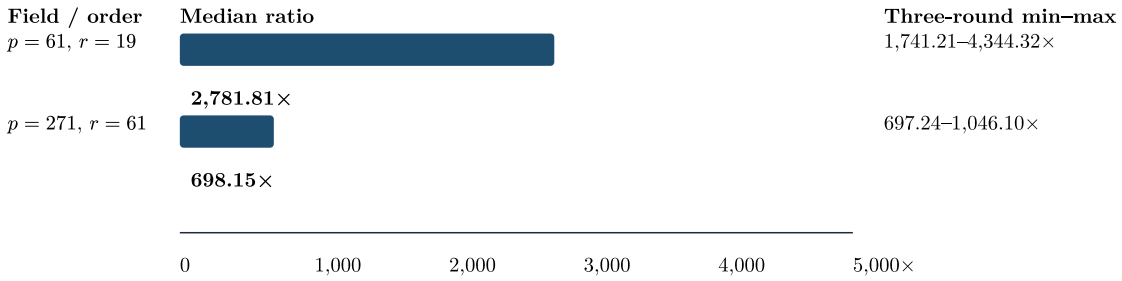
(p, r)	Variant	$S_{\mathrm{CPU/GAE}}$ median [range]	Ratio / floor	Paired / rho
$(61, 19)$	Rho	64.01 [39.01, 98.06]	72.23	1.00
	Rho A/A	50.12 [34.61, 71.05]	56.56	1.11
	Pairing + scalar replay	163,807.34 [161,020.04, 170,683.25]	184,836.79	2,666.48
	Pairing + all checks	170,737.19 [169,493.47, 178,065.56]	192,656.29	2,781.81
$(271, 61)$	Rho	32.76 [28.97, 33.27]	36.97	1.00
	Rho A/A	40.90 [28.42, 74.12]	46.16	1.23
	Pairing + scalar replay	21,978.75 [21,607.04, 28,981.92]	24,800.37	660.71
	Pairing + all checks	23,193.81 [22,872.96, 30,304.74]	26,171.41	698.15

Paired CPU ratio

Cold pairing cost / same-target rho

V2 frozen pairing transfer · 2026-10-15 05:55 UTC · shared Darwin arm64, L0

Each bar is the median of three **paired full current-thread CPU ratios**. The pairing total includes construction, candidate search, both pairings, the multiplicative-field solve, separate source-scalar replay, and the post-solve pairing checks. Blue bar length: 1 inch = 1,000× rho.



Source: `SUMMARY_V2.json` derived from 30 native raw commands in `runs_v2/`, commit `b4e2bff9cb`. Rho A/A and all six independent scalar replays passed. The min–max interval is a three-run range, not a confidence interval. CPU ratios on this shared host are descriptive.

The graph uses `GRAPH_V2.typ` and `SUMMARY_V2.json`. The fully validated row adds three post-solve pairing checks to the complete DLP method. The separate pairing-solve row charges cold construction, unsuccessful search, both pairings, the finite-field solve, and independent source scalar replay. Setup reuse is one target. This is not an amortized result.

Charged phase and primitive ledger

The median current-thread phase times in microseconds are:

(p, r)	H search	Miller	Final exp.	Field DLP	Full total
$(61, 19)$	101,723.959	969.000	759.625	48.917	108,490.417
$(271, 61)$	23,873.166	209.833	143.333	19.334	25,681.333

Candidate search used 26,469 of 27,169 selection-phase extension multiplications at $p = 61$ and 39,115 of 39,673 at $p = 271$. The H-search charge includes the group-order recurrence and cofactor calculation, as well as failed x candidates and projections. Whole-workflow totals including extra checks were 16,491,269 or 5,602,853 base-field multiplications and 46 or 64 base-field inversions. Pairing peak process RSS medians were 6,586,368 and 6,602,752 bytes. Every V2 phase subtotal reconciled with its aggregate; V1 retained the same scalars and deterministic primitive totals. V1 raw files and its own report remain under `runs/` and `reports/run-20261015T0546Z/`.

The measured pairing route solved both frozen public targets, but its cold cost exceeded same-target rho in both cells. H search dominates these two runs. A reusable-H/alpha experiment needs an actual target count and must charge setup once; a warm estimate is not a single-target result. The still-missing nonfactorable chi polynomial decomposition oracle and a different base-field non-isomorphic source/destination pair remain separate unperformed tests. The present degree-three image also has a public scaling isomorphism over these quadratic fields, so this run cannot attribute an isogeny-specific relation effect.

The run used shared-host L0 current-thread CPU. A/A rho varied substantially. Input parsing, output serialization and calibration were excluded from both timed arms. These three-round ranges are descriptive, not confidence intervals or an isolated wall-speed claim. The full source-linked report, run records, exclusions and requirement statuses are in `RESULTS.md`.